

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart Lender-Applicant Credibility Prediction for Loan Approval
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines the scalability strategy and future enhancement roadmap for the **Smart Lender** project. The platform is designed to support increasing numbers of users, larger financial datasets, and advanced AI-driven loan recommendation features. It also highlights the current limitations of the system and the planned improvements to enhance performance, security, scalability, and overall user experience..

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited loan datasets for prediction	May reduce recommendation accuracy for diverse user profiles	High
2	Supports only basic loan recommendation models	Limited personalization for users with complex financial histories	Medium
3	No integration with banking APIs or credit bureaus	Loan eligibility predictions rely on user-provided information only	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users	Deploy cloud-based infrastructure with auto-scaling to support thousands of users simultaneously
2	Data Storage	Uses a local database for storing user information	Migrate to scalable cloud databases (MongoDB Atlas/MySQL Cloud) with backup and replication
3	Performance	Loan prediction and chatbot response times are optimized for small datasets	Implement caching, optimized ML models, and load balancing for faster response
4	Security	Basic authentication and encrypted passwords	Add Multi-Factor Authentication (MFA), OAuth login, end-to-end encryption, and regular security audits

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integration with Banking APIs and Credit Score Services	3 Months	Improves accuracy of loan eligibility prediction
Phase 2	AI-Based Financial Advisor with Personalized Budget Planning	6 Months	Provides intelligent financial guidance and improves user engagement
Phase 3	Mobile Application (Android & iOS)	9 Months	Increases accessibility and user adoption
Phase 4	Multi-Bank Loan Comparison and Real-Time Loan Approval Tracking	12 Months	Enables users to compare offers and track applications efficiently